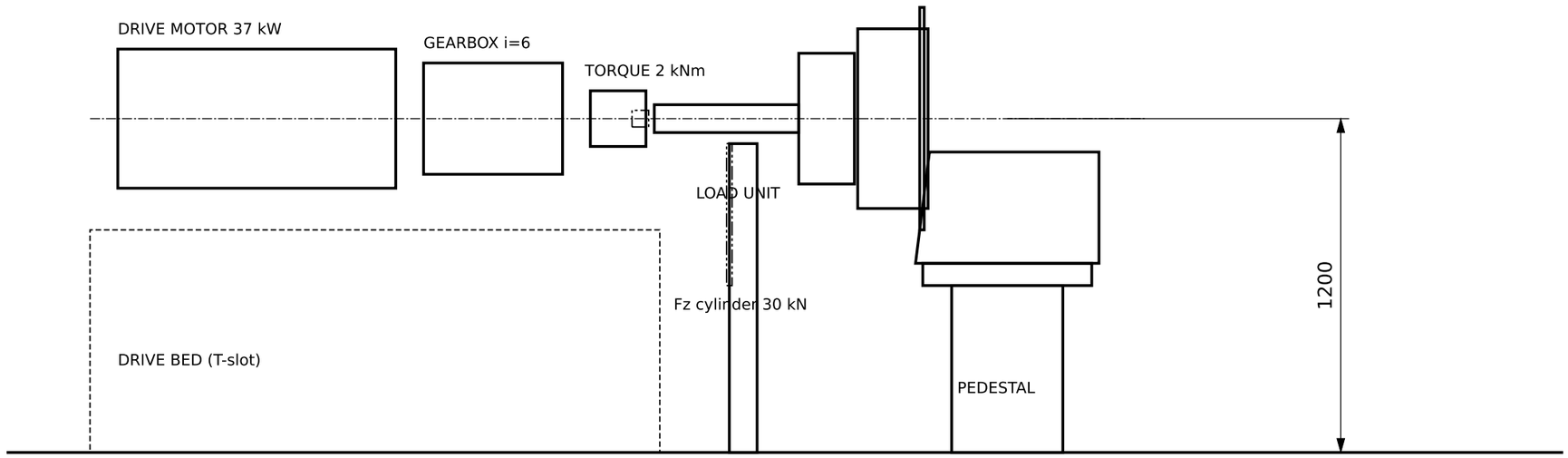
SIDE VIEW · spindle horizontal (5° wedge under the yaw bearing)



NOTES

1. Energy is recirculated: drive inverter and generator-side load inverter share one DC bus; the grid supplies only the losses (≈ 3 kW at rated).
2. Load unit: spherical roller bearing on the dummy hub shaft at x = −200, pushed by a vertical cylinder (≤ 30 kN, servo-hydraulic) and an axial cylinder (≤ 45 kN, static tests only). Couplings each side of the torque transducer decouple bending from the torque reading.
3. Pedestal, load frame and drive bed bolted to a strong floor or a 6 t concrete block; first pedestal mode > 25 Hz (well above 250 rpm × 3 = 12.5 Hz).
4. Guards: full enclosure of the rotating drum, coupling guards, magnet hazard zone during assembly, interlocked access door wired into the bench safety relay.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE (ICELAND) · NACELLE PROTOTYPE P1

BENCH TEST RIG · GENERAL LAYOUT

DRAWING NO.	REV / DATE	SCALE (A3)
NP1-620	A · 2026-11-02	1:20
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see doc 15		Claude / not checked